

Financial ML Track Research Report

1. Problem Framing

The task is to predict whether the next NIFTY 50 close is above the current close. I define the label as 1 when `close[t+1] > close[t]` and 0 otherwise. Flat returns are therefore treated as down/non-up days. I did not remove near-zero moves, because removing them would make the target easier after observing the next-day return. In this dataset there are 1 exactly flat next-day returns and 65 near-zero returns with absolute next-day return below 5 basis points.

The unconditional up-day rate in the usable sample is 53.09%, so a majority-class classifier has a full-sample baseline accuracy of 53.09%. In the locked July-December 2025 test period, the majority-class baseline accuracy is 51.20%. This is the minimum hurdle: a useful model should beat this baseline out of sample and should do so by enough that random variation is an implausible explanation.

2. Data, Split Design, And Tracking

- Total usable observations: 987
- In-sample walk-forward period: January 2022 through June 2025
- Locked out-of-sample period: July 2025 through December 2025
- Expanding walk-forward folds: 36
- Out-of-sample observations: 125

The validation loop trains on months 1-N, predicts month N+1, shifts forward, and repeats. The July-December 2025 block is locked away until the final evaluation. Every training fold is wrapped in `mlflow.start_run()` from the beginning of the fold, with parameters, metrics, prediction artifacts, and model artifacts written to the committed `mlruns/` directory.

3. Feature Audit, EDA, And Final Features

I used exactly 12 features. The selection rule was conservative: use features that can plausibly be known at today's close, are trailing/same-day rather than forward-looking, and are interpretable enough to defend. I did not find an explicit tomorrow-close column. I still dropped opaque engineered columns where the construction was not auditable.

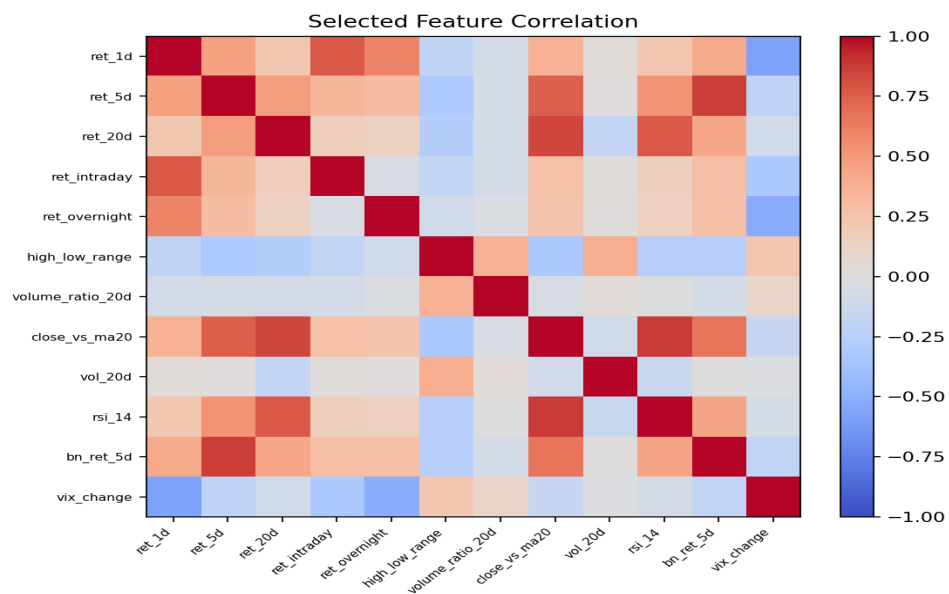
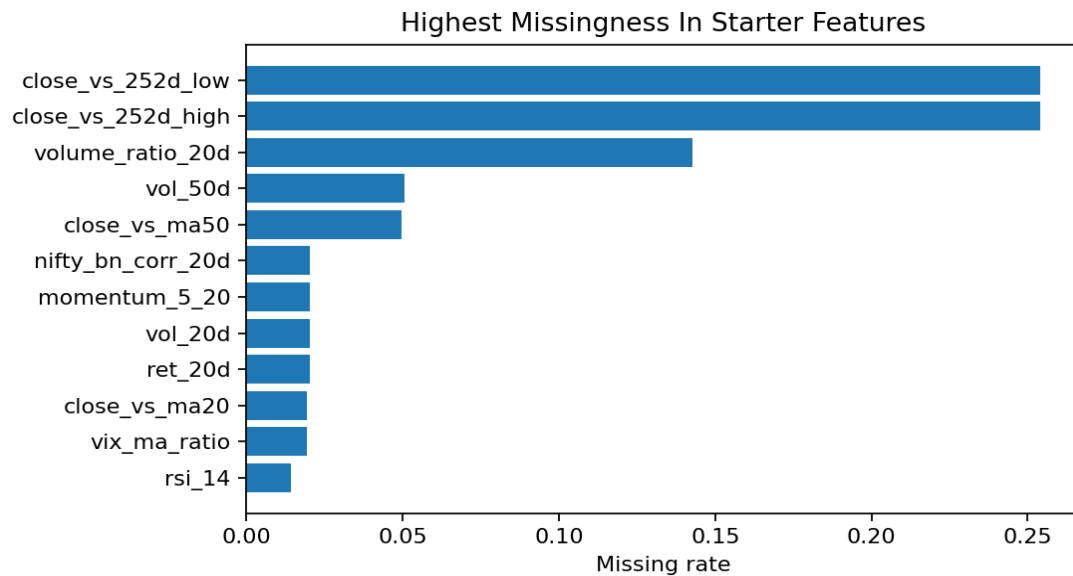
feature	safe_timestamp	justification
ret_1d	T+0 close, after NSE market hours	Captures immediate NIFTY momentum/reversal using only today's close versus yesterday.
ret_5d	T+0 close, after NSE market hours	Measures one-week index momentum while staying fully trailing.
ret_20d	T+0 close, after NSE market hours	Represents roughly one-month trend pressure available at today's close.
ret_intraday	T+0 close, after NSE market hours	Separates same-session open-to-close demand from overnight movement.
ret_overnight	T+0 close, after NSE market hours	Captures gap risk and overnight sentiment visible by today's close.
high_low_range	T+0 close, after NSE market hours	Proxy for same-day realized volatility and uncertainty.
volume_ratio_20d	T+0 close, after NSE market hours	Compares current participation with trailing liquidity conditions.
close_vs_ma20	T+0 close, after NSE market hours	Shows current price extension versus a trailing one-month moving average.
vol_20d	T+0 close, after NSE market hours	Measures trailing one-month realized volatility of NIFTY returns.
rsi_14	T+0 close, after NSE market hours	Standard trailing momentum oscillator for overbought/oversold behavior.
bn_ret_5d	T+0 close, after NSE market hours	Adds related-sector momentum from Bank Nifty over the same recent window.
vix_change	T+0 close, after NSE market hours	Captures same-day change in implied volatility/risk appetite.

All selected features are safely computable only after the current trading session has completed. The intended prediction timestamp is therefore after the T+0 close, for a decision about the T+1 close direction. These features should not be used for an intraday prediction made before the current close is known.

EDA changed three choices:

- `outputs/eda_missingness.png`: the missingness plot showed that `close_vs_252d_high` and `close_vs_252d_low` have about 25% missing values because they need a full one-year lookback, so I dropped them.
- `outputs/eda_correlation_heatmap.png`: the correlation heatmap showed overlapping trend/momentum clusters, so I kept one representative from each group instead of keeping `ret_5d`, `ret_10d`, `ret_20d`, `close_vs_ma5`, `close_vs_ma20`, and `momentum_5_20` all together.
- `outputs/feature_importance.csv`: final-model importance concentrated on volatility, overnight return, and moving-average distance, which confirmed that the selected set still covered volatility, gap risk, trend, liquidity, cross-asset momentum, and VIX after dropping opaque features.

Scaling decision: tree-based LightGBM does not require standard scaling, so I did not standardize features. Missing values are median-imputed inside the sklearn pipeline so imputation is fit only on each training window.



The main suspicious column was `ma5_smooth_signal`. It is not a raw market measurement such as return, volatility, volume, or VIX. It is a prebuilt "signal" column, its exact formula is not documented in the data bundle, and its name suggests a smoothed trading rule rather than a transparent feature. Because the assignment asks for leakage awareness and because the provided starter features were explicitly "not rigorously audited," I treated it as medium leakage/model-design risk and removed it.

Dropped-feature details are written to `outputs/feature_audit.csv`. The key removals were sparse 252-day features, highly overlapping trend/volatility variants, and `ma5_smooth_signal` because it is signal-like and undocumented.

4. Model

The model is a single `lightgbm.LGBMClassifier`. I removed logistic regression, random forest, and sklearn's `GradientBoostingClassifier` so the experiment is not a model-selection exercise disguised as a final result. Hyperparameters are intentionally modest: shallow trees, small leaves, regularization, and subsampling to reduce overfitting on a short financial time series.

5. Results

Walk-forward performance is weak. The model's walk-forward AUC is 0.478, compared with 0.464 for the majority baseline. The model's walk-forward hit rate is 50.34%, while the majority baseline hit rate is 51.15%. In plain English: before the locked test period, the model is not convincingly better than a naive class-prior rule.

Walk-forward fold snapshot; full fold table is in `outputs/walk_forward_folds.csv`:

fold_id	test_start	test_end	hit_rate	auc
1	2022-07-01	2022-07-29	0.2857	0.4388
2	2022-08-01	2022-08-30	0.5500	0.4762
3	2022-09-01	2022-09-30	0.5000	0.4643
34	2025-04-01	2025-04-30	0.4737	0.5952
35	2025-05-02	2025-05-30	0.4762	0.4630
36	2025-06-02	2025-06-30	0.4286	0.4135

In-sample walk-forward metrics with 95% bootstrap confidence intervals:

metric	lightgbm	majority_baseline
auc	0.4776 [0.4428, 0.5194]	0.4644 [0.4254, 0.5100]
balanced_accuracy	0.4846 [0.4543, 0.5158]	0.4843 [0.4568, 0.5172]
fn	139.0000 [120.0000, 159.0000]	107.0000 [88.0000, 126.0000]
fp	228.0000 [201.0000, 252.0000]	254.0000 [229.0000, 282.0000]
hit_rate	0.5034 [0.4723, 0.5386]	0.5115 [0.4777, 0.5501]
max_drawdown	-0.1245 [-0.3895, -0.1101]	-0.2639 [-0.3931, -0.1020]
sharpe_ratio	0.0508 [-0.9980, 1.0804]	0.1196 [-0.9422, 1.3560]
tn	102.0000 [85.4750, 119.0000]	76.0000 [61.0000, 93.0000]
tp	270.0000 [243.4750, 295.5250]	302.0000 [276.0000, 328.0000]
turnover	0.2940 [0.2940, 0.4804]	0.0041 [0.0041, 0.4140]

Locked out-of-sample metrics with 95% bootstrap confidence intervals:

metric	lightgbm	majority_baseline
auc	0.5239 [0.4192, 0.6276]	0.5000 [0.5000, 0.5000]
balanced_accuracy	0.4949 [0.4353, 0.5569]	0.5000 [0.5000, 0.5000]
fn	8.0000 [3.0000, 14.0000]	0.0000 [0.0000, 0.0000]
fp	54.0000 [43.0000, 66.5250]	61.0000 [50.0000, 73.0000]
hit_rate	0.5040 [0.4080, 0.6000]	0.5120 [0.4160, 0.6000]
max_drawdown	-0.0492 [-0.1022, -0.0231]	-0.0454 [-0.1058, -0.0239]
sharpe_ratio	0.7680 [-2.0850, 3.6707]	0.6158 [-2.3653, 3.6332]
tn	7.0000 [2.0000, 12.0000]	0.0000 [0.0000, 0.0000]
tp	56.0000 [46.0000, 68.0000]	64.0000 [52.0000, 75.0000]
turnover	0.1774 [0.1248, 0.3065]	0.0000 [0.0000, 0.0000]

On the locked OOS set, the model hit rate is 50.40%, below the majority baseline's 51.20%. AUC is 0.524, only slightly above 0.50. The model Sharpe is 0.768, but the confidence interval is very wide and includes strongly negative values. That is an important result, not a footnote: it means the apparent OOS trading performance is too uncertain to claim a real economic edge. The baseline Sharpe is 0.616, which is close enough that the model does not clearly dominate a naive long-up-class rule.

Feature importance from the final model:

feature	importance
vol_20d	97
ret_overnight	76
close_vs_ma20	66
volume_ratio_20d	42
ret_1d	40
vix_change	39
high_low_range	35
ret_intraday	33
bn_ret_5d	29
ret_20d	25
rsi_14	15
ret_5d	10

6. Is The Edge Real?

My conclusion is no: there is not enough evidence to claim a real predictive or tradable edge.

First, the OOS directional edge is negative. The model's hit rate is 50.40%, while the majority baseline is 51.20%. A paired permutation test of model hit-rate edge versus the majority baseline gives $p=1.0000$. This does not reject the null that the apparent difference is noise.

Second, the random-label test does not support a robust signal. Keeping the model scores fixed and shuffling OOS labels gives an AUC p-value of 0.3246. A result this close to random labels is not strong evidence of learnable structure.

Third, the train/test boundary sensitivity is unstable. If a small change in the OOS start date changes the conclusion, the result is probably regime- and sample-dependent rather than a durable edge.

scenario	train_end	test_start	test_end	test_rows	model_auc	baseline_auc	model_hit_rate	baseline_hit_rate	model_sharpe	baseline_sharpe
oos_from_2025_06	2025-05-31	2025-06-01	2025-12-31	146	0.4978	0.5000	0.5068	0.5274	0.1135	1.1912
oos_from_2025_07	2025-06-30	2025-07-01	2025-12-31	125	0.5239	0.5000	0.5040	0.5120	0.7680	0.6158
oos_from_2025_08	2025-07-31	2025-08-01	2025-12-31	102	0.5412	0.5000	0.5392	0.5392	1.6862	1.9520

Fourth, the OOS Sharpe ratio looks superficially positive, but its 95% bootstrap interval is wide: -2.0850 to 3.6707. A real edge should survive uncertainty estimates; this one does not. This is especially important because the simple backtest ignores transaction costs, slippage, financing, margin, and operational constraints.

Diagnostics:

test	statistic	value	interpretation
hit_rate_vs_majority_baseline	two_sided_p_value	0.8580	No statistically reliable directional edge if p-value is above 0.05.
paired_permutation_edge	observed_hit_rate_edge	-0.0080	Model hit rate minus majority-baseline hit rate on the same OOS dates.
paired_permutation_edge	p_value	1.0000	Probability of an absolute edge this large under paired sign randomization.
random_label_baseline	observed_auc	0.5239	Observed model AUC on the true OOS labels.
random_label_baseline	random_auc_mean	0.4999	Mean AUC after shuffling OOS labels against fixed model scores.
random_label_baseline	auc_p_value	0.3246	Share of random-label trials with AUC at least as high as observed.

7. What Could Still Be Wrong?

Several things could still be wrong even after the leakage audit. The starter features were provided as a convenience and not fully documented, so some rolling calculations may have implementation assumptions that are not visible from the column names. Same-day close-based features are acceptable only if the trading decision is made after the close for the next session; they would be invalid for an intraday signal. The bootstrap intervals treat resampled days as independent, which is imperfect for financial time series. The strategy diagnostics use a simple long/short direction rule and do not include transaction costs. Finally, the sample is short and covers a limited market regime, so the model may be fitting noise.

8. What I Would Build Next

The next version should focus less on model complexity and more on validation quality. I would add transaction-cost-aware backtests, block bootstrap confidence intervals, probability calibration, threshold tuning using only walk-forward folds, and a true external test period when more data becomes available. I would also rebuild all features from raw OHLCV instead of relying on precomputed starter columns, then compare a no-trade option for low-confidence days against the always-long/always-short simplification used here.

Bottom Line

This project produced a reproducible MLflow-tracked research pipeline, but the honest research conclusion is negative. The LightGBM model does not beat the majority baseline on locked OOS hit rate, its walk-forward performance is near random, and its positive OOS Sharpe is too uncertain to defend as evidence of an edge.